An important principle of cryptography, known as *Kerckhoff's Principle*, says that an encryption scheme should be secure even if every aspect of the algorithm is known, so long as the key remains secret.

> **PRINCIPLE** You should use only algorithms that have been designed through an open process with public review by experts, such as the algorithms you'll use in this chapter.

There are several secure encryption algorithms in current use, but the most important is the *Advanced Encryption Standard*, AES, which was standardized in 2001 after an international competition, and is widely considered to be very secure. AES is an example of a *block cipher*, which takes a fixed size input of 16 bytes and produces a 16-byte encrypted output. AES keys are either 128 bits, 192 bits, or 256 bits in size. To encrypt more (or less) than 16 bytes with AES you use a block cipher *mode of operation*. The choice of mode of operation is crucial to the security as demonstrated in figure 6.4, which shows an image of a penguin encrypted with the same AES key but with two different modes of operation.[1] The Electronic Code Book (ECB) mode is completely insecure and leaks a lot of details about the image, while the more secure Counter Mode (CTR) eliminates any details and looks like random noise.

> **DEFINITION** A ***block cipher*** encrypts a fixed size block of input to produce a block of output. The ***AES*** block cipher operates on 16-byte blocks. A block cipher ***mode of operation*** allows a fixed-size block cipher to be used to encrypt messages of any length.